

ASSIGNMENT 1-REFLECTIVE REPORT



SMART IRRIGATION SYESTEM

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1. Introduction

This piece of self-reflection presents and documents the process of implementation of my thesis project from research to the development of functional artifact, which was focused on creating Smart Irrigation System with IoT and AI. The main objective of the project was to come up with a system for irrigation which is smart, automated and able to adjust the water consumption according to the available real-time environmental data, thereby solving the major agricultural issues related to water scarcity and the inefficient use of irrigation together with the resource management. By using the irrigation system, farmers can remotely monitor and manage the irrigation process, which is based on AI-powered predictions to provide the most precise water distribution.

In this report, I will discuss on the various approaches that I took for the successful and timely completion of my project. Discuss how effective time management and proper planning helped me to stay on track. Proper planning was the key aspect as the idea was new to me, and I have never implemented something like this. I will also discuss the fallbacks or the obstacles I faced, some aspects that I could not accomplish. I shall also discuss the limitations or the areas for future improvements in the project. This shall highlight potential enhancements to the project.

I shall also reflect on various other aspects like problems solving approaches I took to overcome certain challenges faced, although I could not solve all of them. I will also discuss on the interactions with my supervisor, how supportive and helpful he was, and how vital his assistance was for the completion of my artifact. I shall also perform some assessment of the completed work and highlight some of its limitations. And finally, I shall include the most important part, the personal growth and development of my knowledge, skill and experience throughout the journey from research to the development process of Smart Irrigation System.

2. Motivation

I am grown up in an agricultural family, witnessing the difficulties of manual irrigation. The practice of supplying water and stopping it at the right time was not only a lot of tedious work but also was very inaccurate. Different crops need water in different amounts at different stages of their life, which made it hard to keep track of all the data and control the process manually. This commonly resulted in too much irrigation, which was water being wasted, or the plants not getting enough water, which affected the yield adversely.

Having to observe such inefficiencies of manual irrigation closely in everyday life, I was motivated to create a smart and automated system that could optimize the water usage by delivering the exact amount of water to the specific plants as per its water requirement while also considering the environmental factors like temperature and humidity. And also as a computer science student introducing the modern technologies in the traditional farming was always on my to-do list. So I

decided to create a smart, efficient and practical solution to the aforementioned problem which not only ease the farmers workload but also optimize the resource utilization and improve productivity.

3. Time Management

Time management was one the most crucial part for the successful completion of the project. Since the concept of IoT is a bit new to me, and also integrating it with remote server and the AI model is a complex task. It required thorough technical knowledge and deep research. I simultaneously needed to be involved in literature review, system design, implementation, testing, and documentation. To keep up my focus, I planned out my work by first clearly setting out the milestones and deadlines whereby I was able to split my workload into simpler tasks.

Despite this, self-discipline and following my schedule at proper times were very challenging. There were moments when I stumbled upon unexpected hurdles as integrating the IoT components, or refining AI predictions thus requiring more time and effort leading to the delays in my original plan. Still, I was resilient enough to adjust my schedule in accordance with the problems to make it possible to achieve the desired progress.

In order to better optimize my time management, I consulted my supervisor Mr. Krishna Aryal. And after the meaningful discussions, we came up with the following key strategies that I strictly need to follow.

- **Define Clear Goals:** Create a detailed roadmap of the project, split the tasks into smaller chunks. It helps to maintain the clarity and focus.
- **Setting Priorities:** Identify the most critical aspects of the system like fine tuning the AI model, calibrating the sensor components etc, and address them first.
- **Progress Tracking:** I kept the track of tasks completed in each day, and which tasks are to be completed afterwards. Also the tasks which need the further refinement were highlighted. And also the Problems were pinpointed.
- **Iterative Development:** As per my supervisor's suggestion, rather than trying to develop and perfect a feature in one go, I adopted the step by step approach. Implement a basic feature at first, then keep on improving it based on testing and feedback.
- **Immediate Problem Solving:** When new problem or issue arose, I sought for immediate solution, try to solve it as fast as possible, rather than scheduling it for later which further delays the progress.

By following these strategies, I was not only productive but also able to meet by key milestones in the desired time frame. Besides, for the better visualization and tracking of the progress, I created GANTT charts which broke down the whole projects into smaller milestones and help me better organize my time. Following are the images of my projects Gantt charts which shows clear road- map of smart irrigation system development.

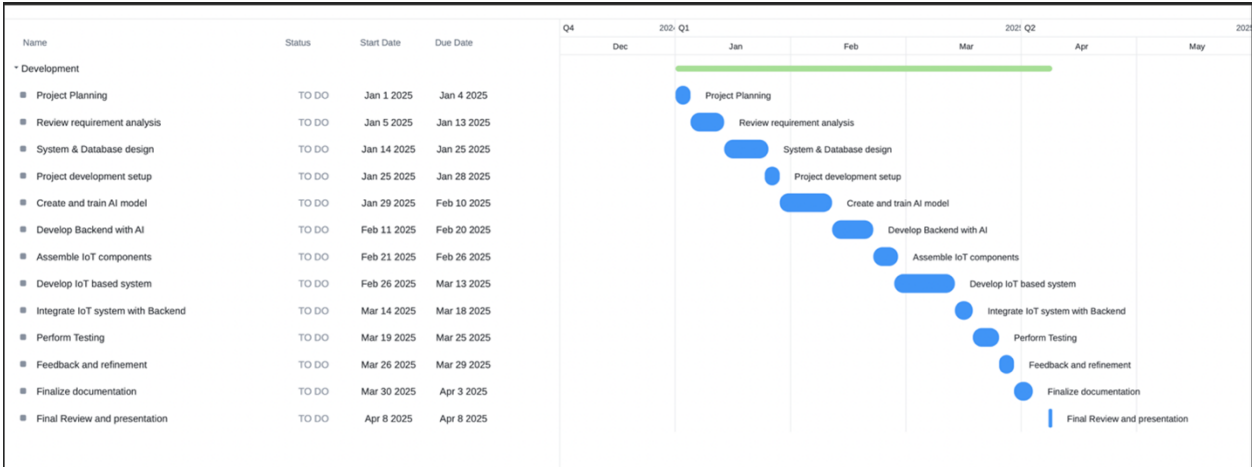


Figure 1:Initial Gantt chart

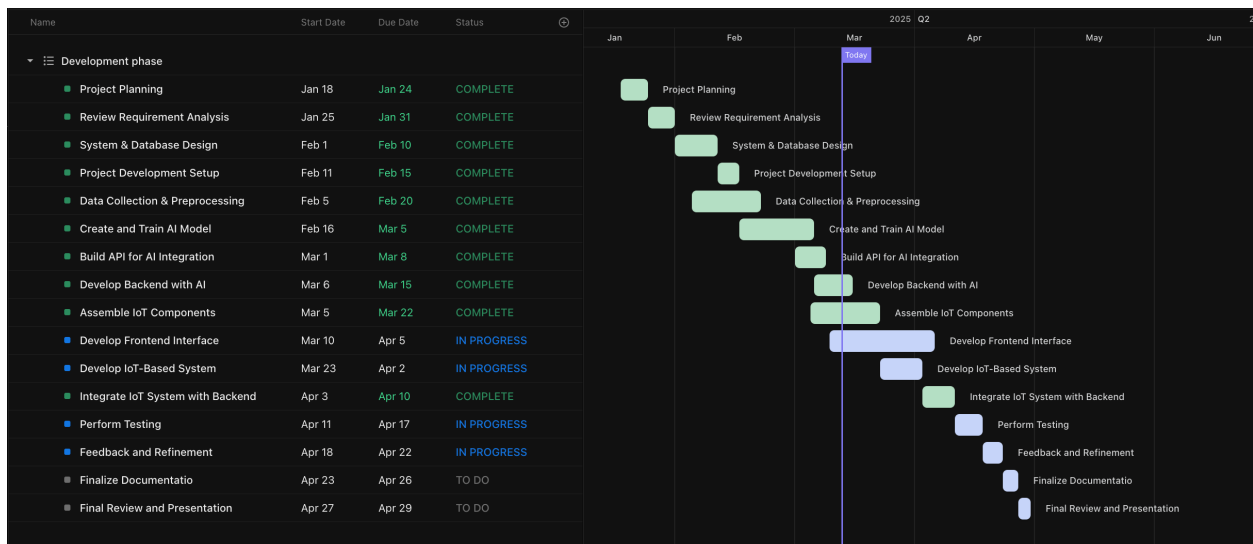


Figure 2:Revised Gantt chart

4. Evaluation

4.1 Communication with supervisor

The continuous conversations that I had with my supervisor Mr. Krishna Aryal has been one of the main contributing factors that has shaped the project toward the right direction and also helped me refine the approaches I take to accomplish a task. The regular meetings were my source of valuable information that helped me stay on the track and at the same time, tackle and overcome the challenges in time.

One of the most significant subject we talked about was optimization of the development process. I initially chose to complete the features in a linear fashion, perfect the feature in one go, then move

to the next one. And it was taking a lot of time for me to actually perfect the feature as I was new to the IoT development. Then my supervisor advised me to adopt iterative development process. Starting with a basic implementation and refining it step by step through testing and feedback. This approach significantly improved the development speed, as now I was not stuck at perfecting the one feature, rather once a basic implementation is complete, I would move on to the next, and later refine it.

The other issues were the technical constraints of AI model integration, server-based crop tracking, and remote monitoring technology. I came up with the idea of using a RTC module for tracking the crop age in the IoT device itself. But after a while, my advisor and I decided that using a server based crop days tracking solution would be much better as it would remove the need of extra hardware, and also make the project more scalable and affordable.

He also emphasized on the usability of the system. He advised me to design a user-friendly interface that would be easy for general farmers to use. Hence, I paid attention to making an easy-to-use front end for the online user interface, to track and control the irrigation schedules.

The regular feedback that I got from my supervisor not only helped me to refine the technical approach but also the project's scope, so that the system was practical, efficient, and aligned with the real world agricultural needs.

5. Changes of Plans

Initially, it was my plan to use an RTC (Real-Time Clock) module to monitor crop age, updating it each day by sending requests to the remote server. However, because of my limited knowledge of IoT development, I faced several technical difficulties. Most of the RTC libraries I have found were developed especially for Arduino-based devices, while my system was designed around an ESP-WROOM-32 microcontroller, and this made it particularly hard to find a suitable RTC library.

On top of that, even during the initial setup, the ESP-WROOM-32 board was not directly accessible from the development IDE, therefore I was forced to use the ESP32Dev board instead. This, in turn, caused compatibility issues because some of the hardware components were not working as a result the development process was further delayed. Thus, I was stuck and could not proceed effectively.

When I talked to my supervisor about the problem, he recommended setting up a server-side crop tracking solution instead of RTC hardware. This change was more effective and scalable and has become a preferable option. The ESP32 device is now sending an "update crop age" request to the server periodically. The server checks the last update timestamp, and if at least one whole day has passed, it will increase the crop age accordingly. This method not only resolved the incompatibility issue but also made the development easier and improved the reliability and affordability by removing unnecessary hardware component.

After this insightful outcome from the discussion with my supervisor, I was finally able to move forward with the development process.

6. Analysis of Development Process & Artifact

The development of smart irrigation system was an iterative process, and it was development in different phases step by step. The development phases involved AI model training, backend development, IoT programming, and UI development. Every stage of the process required planning, testing, and debugging to ensure the entire system's performance, reliability and real-world usability of the application. The first stage was focused on the AI model training, which was followed by server-side development along with user interface design and then later IoT device programming. The process turned out to be a tough but at the same time very satisfying one, as different challenges and hurdles were encountered and were gradually solved. Along the journey, I would frequently seek advice from my supervisor, who regularly provided me with the possible architecture, AI model selection, and IoT implementation as well as insightful suggestions when I stumble upon any problem.

6.1 AI Model Training and Integration

One of the earliest and the most important stage of the project was training the AI model to accurately predict irrigation. For the model to work, environmental data such as temperature, humidity, soil moisture, crop type, and crop age were fed to it, which then predicted if irrigation was required. I tried out machine learning algorithms like Random Forest, tensorflow, and MLP Classifier, and finally decided to use Random Forest model which is very lightweight and gave the best accuracy of 98%. hence best fit for a resource-constrained environment.

For the integration to be simplified, I developed a simple python web server, that accepts request via REST API, and returns its decision i.e whether to irrigate or not. (0 for no irrigation, 1 for irrigation). The api can be used by the nodejs server to send the crop information and environmental data received from IoT device to make prediction and receive the decision from AI model.

6.2 Backend Development & Server Communication

The system's backend was developed using Node.js and Python, which is designed to be a scalable and efficient architecture. User and IoT device registrations, authentication of devices, creation of crop session cycles, and monitoring systems and environmental data are handled by the Node.js server, while the Python AI server is responsible for irrigation predictions. To establish the seamless interoperability between all three systems, I designed API driven architecture, where each system communicate with each other via REST API using HTTP requests.

Maintaining the real-time data synchronization between IoT system and remote server was challenging. To solve this, I implemented the use of asynchronous

request handling and logging in a structured database to keep track of irrigation decisions and environmental changes. In addition, the system also records water usage statistics, which it does by determining the total water consumption per crop through water flow rate and duration of irrigation.

6.3 IoT system development & deployment

The IoT irrigation system is developed using ESP32 microcontroller device. It consists of peripheral sensor devices like temperature, humidity and soil moisture sensor to collect environmental data. It has a motor pump attached to it, which it controls automatically to start or stop the irrigation. It required extensive testing and optimization to ensure proper functioning and reliable communication with the remote server. The device periodically sends request to the remote node.js server along with environmental data, and receives the decision whether to irrigate or not, and based on the decision from server, it controls the motor pump to enable or disable irrigation process. The device also sends the live environmental status periodically to the server. Which can be viewed by the user.

To ensure the ease of use, I designed the configuration system to be as simple as possible.

- User registers on the server, then creates a crop session/cycle by entering type of crop and also it's age if already planted.
- User then registers the IoT device to the system providing a name and it's pump flow rate. After this the server generates and provides a device identifier.
- User then turns on the IoT system, connects to it's local WiFi access point, and navigates to the setup page.
- User enters wifi credentials, server address and device identifier.
- Once connected the IoT system now communicates periodically with the remote server sending the environmental conditions and based on the irrigation decision made by server, the system controls the irrigation in the field.
- User can now login to server, and view live system status like environmental conditions, irrigation status, and irrigation schedules along with water usage statistics.

Via web based interface, the server also allows the user to manually start or stop the irrigation.

6.4 User interface and Visualization

I developed a simple and intuitive web app that allows users to track real-time irrigation status, the environmental data as well as view irrigation schedules. Users can also manually start or stop the irrigation via the web app. The UI was designed with user friendliness in mind, so that even the farmers with limited technical experience can use it effortlessly.

Besides, I also developed a water usage tracking feature, which enables users to see the water consumption of a plant throughout it's life. This enables data- driven decision-making for farmers which allows them to efficiently plan which crops to plant based on the season and water availability.

System Name

Farmland System

Edit

Pump Flow Rate

32

Submit

Delete

Identifier:

0194f98e-7018-7554-82ce-f133f9dc72ad



Manage Sessions

Live Status

View Statistics

Preferences

Figure 3:Registration System

Search Crop Sessions



Sugarcane

Potato

Pulse

Groundnuts

Add Crop Session

Pulse

Age Count

50

Edit

Update

Deactivate

Delete

Figure 4:Currently active Session

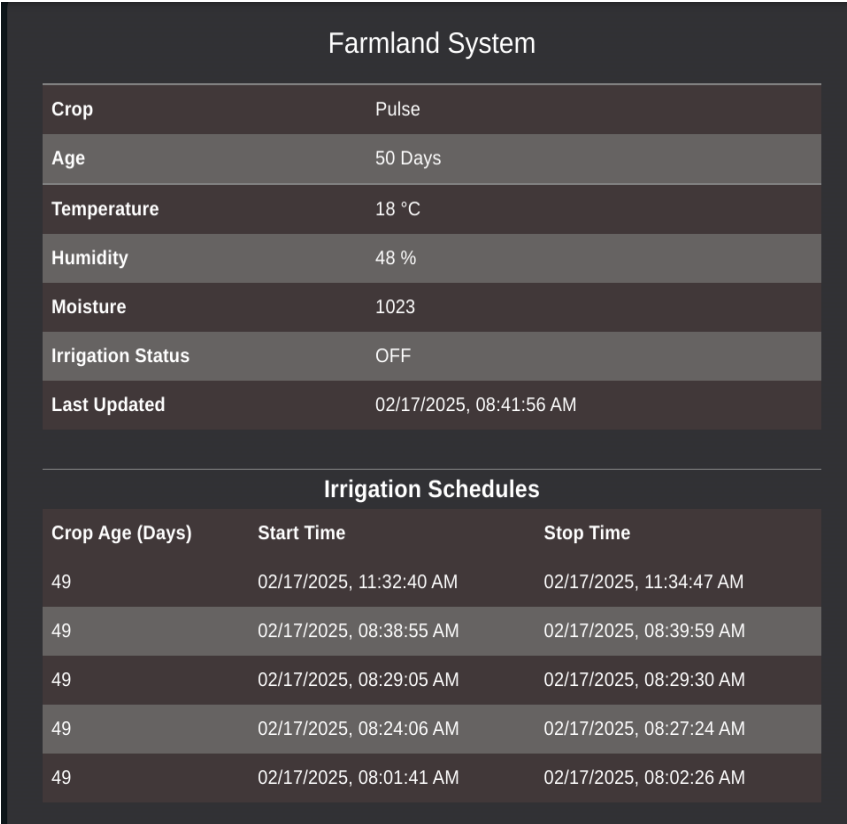


Figure 5: Live Status of Farmland system

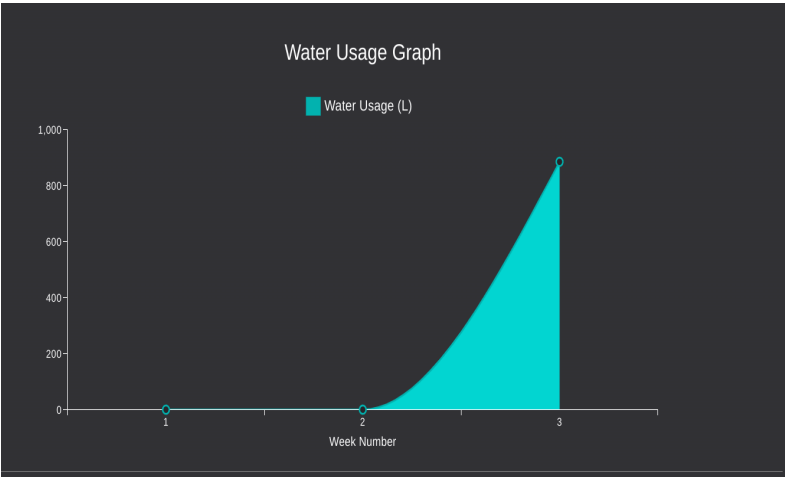


Figure 6: Water Usage Graph by the plant every week

6.5 Final Evaluation and Completion

At this time, the smart irrigation system is completed and fully functional, it uses AI based decision making, IoT automation, and real-time environmental monitoring. The iterative development cycle

allowed me to experiment, refine, and optimize every part, which resulted in a functional and scalable system.

Despite a few struggles like connectivity issues and limited datasets, the project seems to be successful in creating a base to move forward with future improvements.

7. Limitations & Future Enhancements

The process of creating the smart irrigation system was a real challenge for me, however those hindrances encountered shaped the final project outcome as well as my learning experience. Though the project is successful in delivering the core objectives, there is always some room for improvement. One of the major drawbacks was the need for internet connection, which might make it impossible to use a machine in the regions that have a limited or no connection. On top of that, the AI model could still be made more accurate and generalized by adding more diverse datasets and crop types, for now it only supports limited crop types.

Despite the system being functional for now, there are still numerous areas for enhancements. One of the most important ideas that I have taken from this project is that no system is ever truly be finished, it can always be improved and extended. In the coming days, I would love to use other communication methods like LoRa (Long Range) networks to reduce the dependency on the WiFi network. Also, making the system offline could give it more strength in areas where there is no network connectivity.

Another area of improvement is to improve the adaptability abilities to the AI model by putting in real-time weather forecasting and increasing the training dataset to get the better prediction accuracy of the diverse climates and crops. I also agreed that better user experience is needed. Despite my efforts to create the best I can, simplifying the setup process and providing more user-friendly interfaces would make it accessible to a wider audience, including farmers with limited technical knowledge.

Beyond automating the irrigation in an open field where the external factors are uncontrollable. The system can be tailored to monitor and control a greenhouse. Various aspects like fertilizer levels, soil PH, which is the soil's acidity level, light, temperature, and humidity will be monitored in order to provide ideal conditions for the growth of a variety of crops. As a result, precision, climate- controlled farming will be possible, making it feasible to have the crops grown efficiently in any region regardless of external weather conditions.\

Being able to take the project and integrate IoT-based smart irrigation has extended my knowledge of both the possibilities and the constraints of IoT- based smart irrigation. Moving forward, I am happy about the promising prospects of scaling the system and thus improving this system by making it more efficient, cheaper, and applicable to the local and maybe potentially global agricultural communities.

8. Critical Analysis & Reflection

The first stage of my thesis project was to choose the right topic which had to be new and relevant. I was looking for a topic that dealt with new technologies and I especially wanted to use Artificial Intelligence (AI) and the Internet of Things (IoT). I was thinking to develop something like smart home devices combining both AI and IoT however my past experiences and observations in farming sector motivated me to pick something that would not only address the irrigational inefficiencies but also introduce modernization in the agriculture. As a result, the smart irrigation system concept was derived, which uses AI for predictive decision-making and IoT for automated irrigation control.

Throughout the project development journey I dived into several domains, mainly machine learning, IoT based automation, server-client communication, and embedded systems programming. In the process I needed to conduct a deep and fruitful research on how AI models communicate with real-time IoT data and how to establish communication between different servers and systems. The most difficult and yet rewarding part was making both Node.js backend server and the Python AI server work without any problems, so the exchange of real-time data and predictive analysis could be done.

Although I already had some experience and knowledge related to AI and backend development, working with IoT devices and embedded programming was new to me. Knowing the communication protocols, working with hardware constraints, managing limited memory of the ESP32 microcontroller so that it can efficiently work with all the peripheral sensors and devices while also maintaining a reliable communication with the remote server was really challenging. Also ensuring the remote control functionality while the IoT device runs autonomously in the field collecting the environmental data and sending them to the server for prediction and then executing irrigation task was really tough. Furthermore, the AI processing part needed some data preprocessing including optimization in order to get the correct prediction of the irrigation depending on the environmental data. Besides developing a intuitive user interface and data visualization part in the UI making it easier for farmers to remotely monitor the irrigation activity and water usage statistics added another layer of complexity. However at the end the overall system was implemented as I consulted my supervisor at every steps of the development phase. The entire process turned out to be very rewarding as I got to learn many things from this experience.

Finally the project is completed in the limited time also with the limited hardware resource to optimize the affordability. Though the project still has some limitations and flaws, I still call this project a success because the journey of developing this solution has provided me with a very enriching experience. The project was about joining hardware, software, and AI into a system that was practical in the real world. The project got me to work on my problem-solving abilities, helped me to get to a deeper level of understanding of IoT and AI, and made me familiar with the complexities involved in real-world implementation.

9. Thesis Table of Contents

I. Introduction

- Overview of Smart Irrigation System.
- Background and purpose of the project; Emphasize the need of modernization in irrigation, briefly discuss the relevance of AI and IoT in farming.

II. Literature Review

- Briefly discuss about the existing smart irrigation technologies, highlight and summarize their key findings and takeaways.
- Basically, this review presents a concentrated focus on the smart irrigation systems and different approaches of automating the irrigation along with various relevant machine learning algorithms.

III. Artefact Design and Methodology

- Detailed system description. Hardware requirements for hosting the server and also required IoT components.
- Discuss how the overall system is designed and how it works, how IoT, the server & AI model integrate with each other.

IV. Development and Testing

- Documentation of the artifact development including the development of AI model, backend server and IoT devices programming.
- Documentation of the testing and evaluation part.

V. Results and Discussion

- Analysis of the artifact's performance, accuracy metrics and user interactions, and analysis of the system limitations.

VI. Assessment and Critical Analysis

- Evaluation and comparison against original aims and objectives, strengths and weaknesses of the artifact and improvements.

VII. Conclusion and Future Work

- Summary of the key findings, impact and contribution of the system, and areas for the further development and commercialization of the system.

VIII. Conclusion

- Final conclusion of the Smart Irrigation System, how transformative it is. Discuss the knowledge and skills gained throughout the development.

10. Conclusion

This project is no doubt a wonderful and rewarding journey. It is what has helped me strengthen my research and technical skills. Especially, through this process, I have not only gained knowledge in AI, IoT, and system development but also had practical experience in problem-solving, adaptability, and interdisciplinary collaboration. Mistakes I made, whether in training AI models, handling IoT constraints, or ensuring system interoperability, have definitely been the factors influencing my transformation into a developer, researcher and eventually into an engineer.

Besides, the entire process, from research, testing and development to fine-tuning, was made much more interesting by the supervisor discussions and interactions with my friends. Along these path, my supervisor and friends gave me the needed insights and feedback that was not only positive but also constructive, and encouraging which made this project not only technically successful but also a very memorably and fulfilling experience. This unforgettable journey has left me with everlasting memories of my college life and teamwork with my friends. This greatly reinforced my enthusiasm and longingness for future explorations in the realm of AI and IoT automation for modernizing the agriculture.